

A Offline training produces three fixed components

Normal autoencoder

Train on NSRDB ECG1.
Select the checkpoint on other subjects.
Freeze its weights before MLII modelling.

Two Extra Trees models

Use the same 242 MLII features.
Select on earlier validation beats.
Refit the binary gate and subtype model.

Saved models and decision threshold

Frozen autoencoder + binary gate + subtype classifier
Validation-selected abnormality threshold: 0.570833

B Frozen model use

Retrospective temporal test

Later MITDB beats from known patients
Compare predictions with expert labels.

Session prediction

Incoming MLII signal or database replay
Return N or one of five arrhythmia groups.